

Teaching and Learning Mathematics at NUS

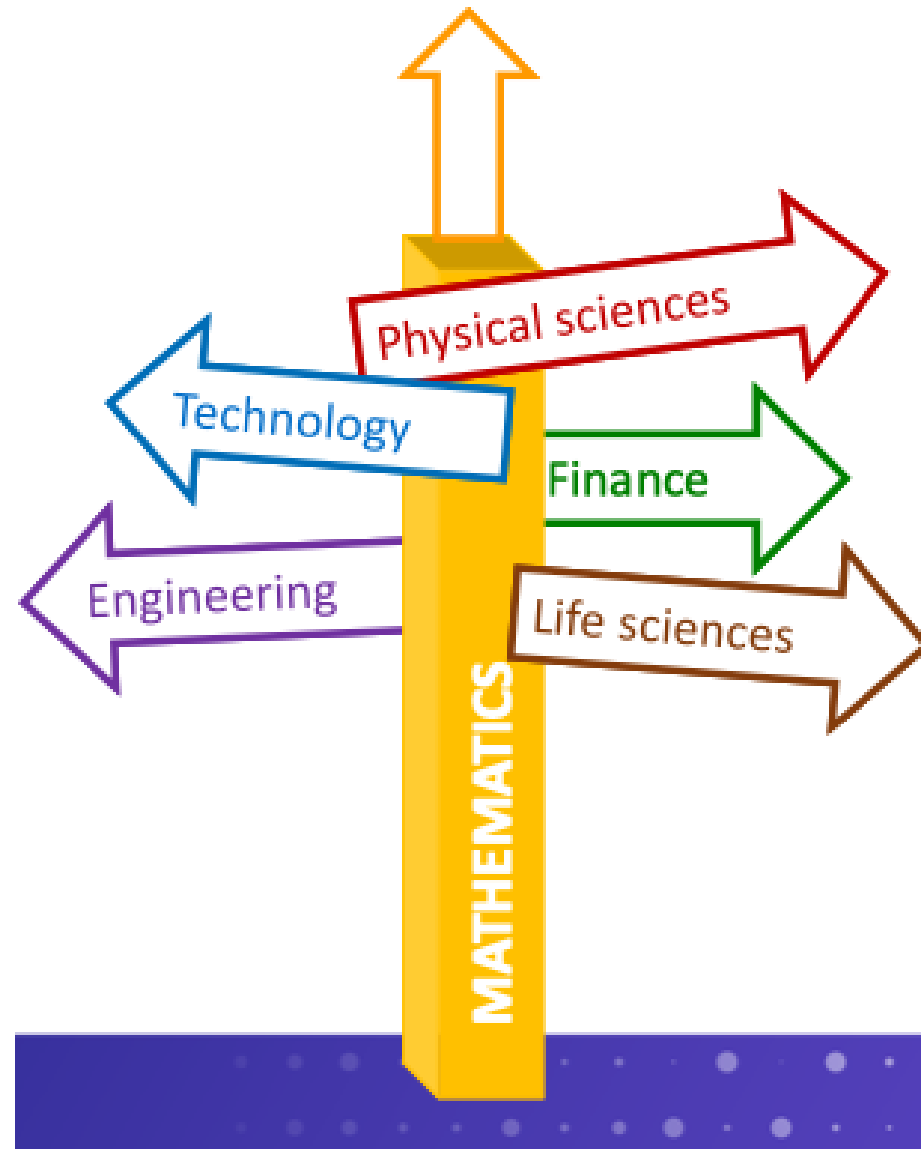
Tân Nguyễn

Hội Thảo Dạy và Học Toán trong Kỷ Nguyên
Công Nghệ



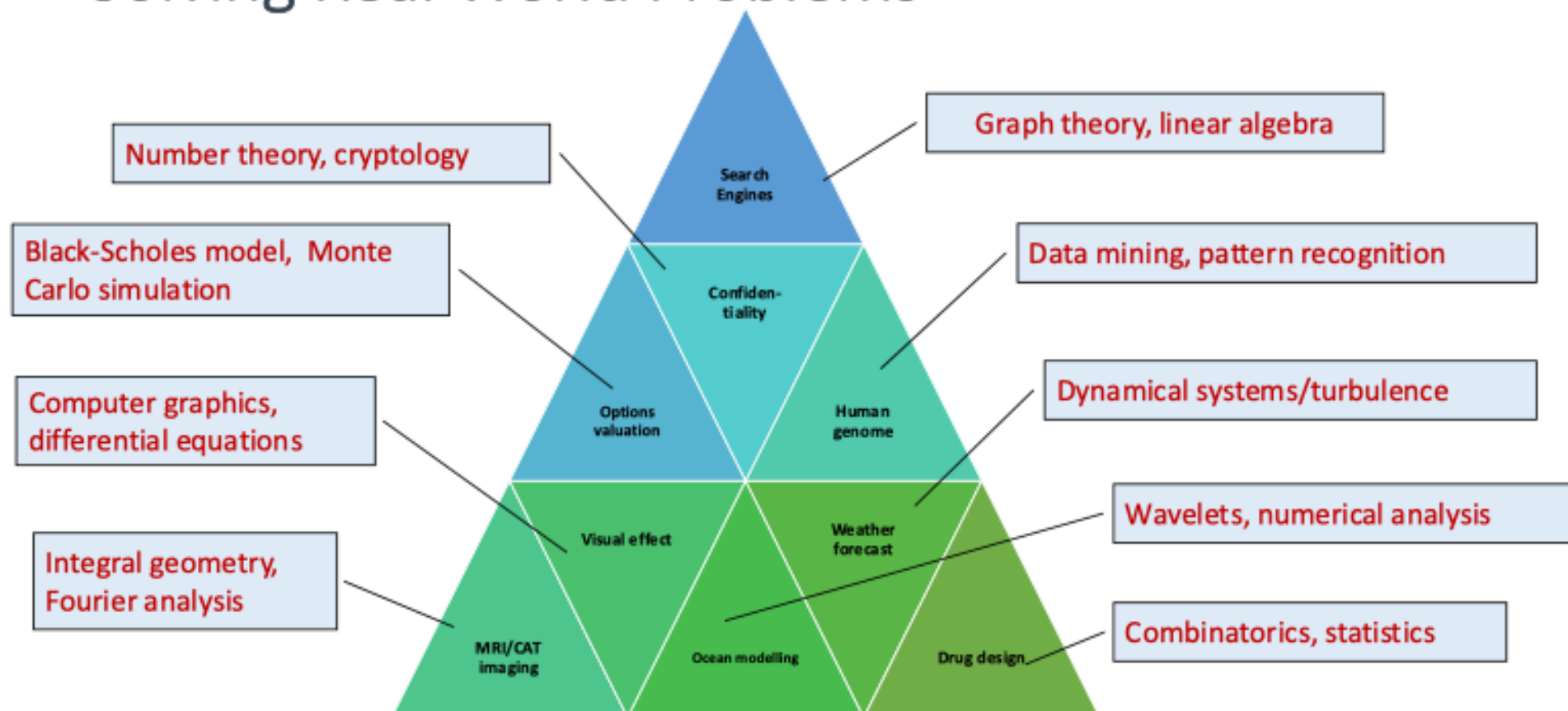
Why Study Mathematics?

Paving the Ways



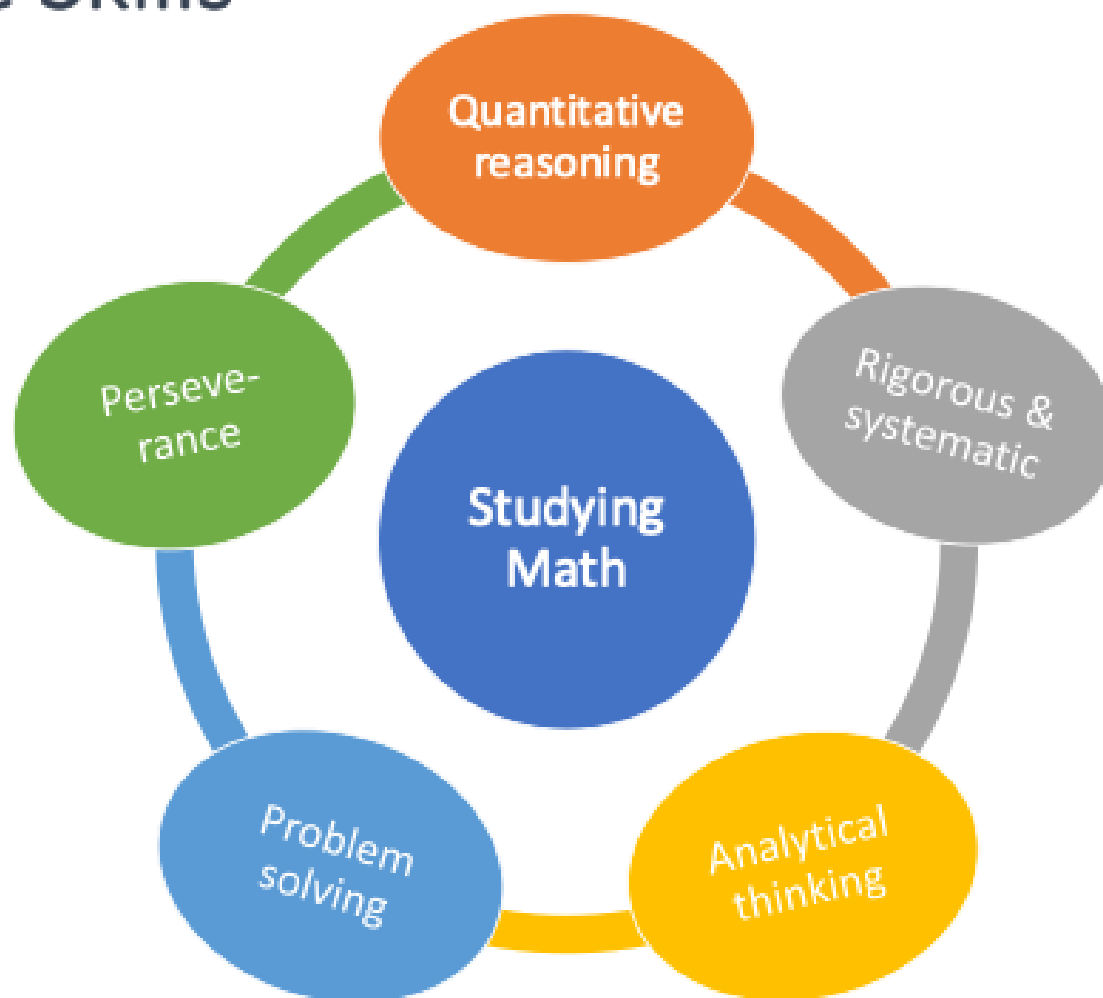
Myriads of Real World Applications

Solving Real World Problems

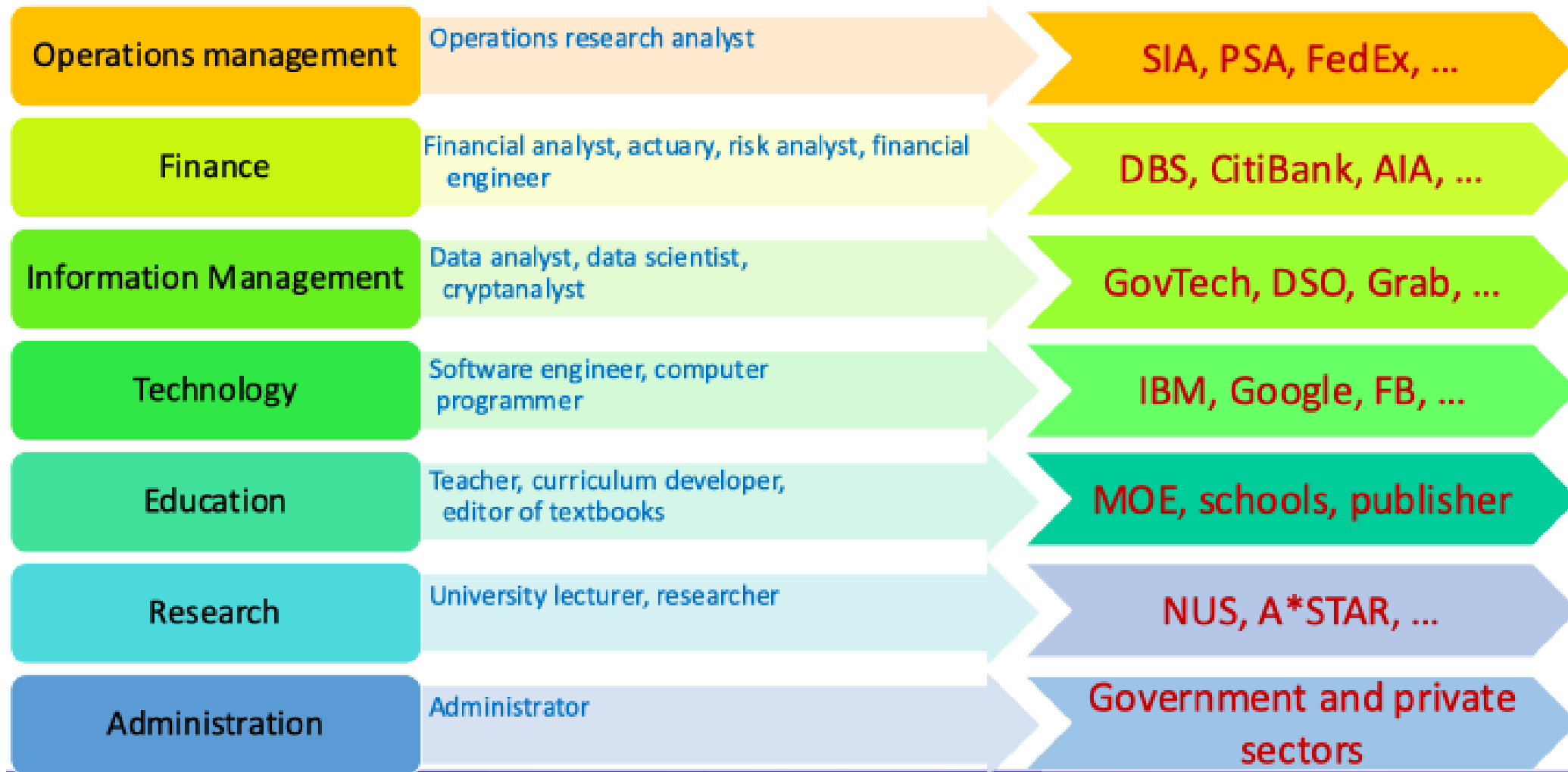


Lifelong Learning Skills

Intangible Skills



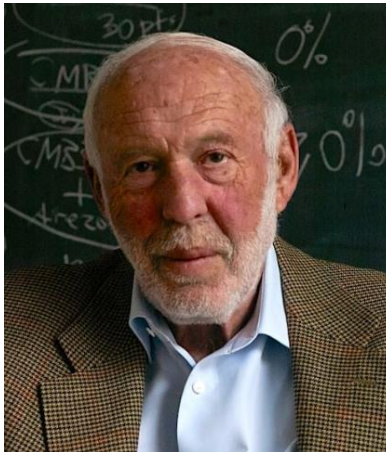
Preparing for Different Career Choices



Successful People with Math Degree



Steve Ballmer
Microsoft
Former CEO



Jim Simons
**Renaissance
Technologies**
Founder



Lee Hsien Loong
Senior Minister
of **Singapore**



Nicușor Dan
President of
Romania

Successful People with Math Degree



GS. Tạ Quang Bửu
Bộ trưởng Bộ Đại
Học và Trung Học
Chuyên Nghiệp
Bộ trưởng Bộ Quốc
Phòng



Trương Gia Bình
Chủ tịch hội đồng
quản trị của **Tập
đoàn FPT**



Trần Đình Long
Chủ tịch Hội đồng
quản trị Công ty cổ
phần **Tập đoàn Hoà
Phát**



Phạm Nhật Vượng
Sáng lập và Chủ
tịch **Tập đoàn
Vingroup**

Mathematics Programme in NUS

Mathematics Programme

- 4-year honours degree programme
- Offers a wide range of pure and applied math modules and a broad spectrum of mathematical research activities
- Includes the study of mathematical methods and problem-solving techniques that are applied in science and other disciplines
- 3 Specializations (optional):
 - a. Data Modelling and Analytics
 - b. Operations Research and Analytics
 - c. Pure Mathematics

Math Module Requirement

1 Gateway Module

- MA1100 Basic Discrete Mathematics
- MA1100T Basic Discrete Mathematic

6 Essential Modules

- MA2001 Linear Algebra I
- MA2002 Calculus
- MA2101 Linear Algebra II
- MA2104 Multivariable Calculus
- MA2108 Mathematical Analysis I
- MA2116 Probability

1 Project Module

- MA4198 Capstone Project in Mathematics

7 Elective Modules (sample)

- MA3206 Set Theory
- MA3210 Mathematical Analysis II
- MA3211 Complex Analysis I
- MA3218 Applied Algebra
- MA3220 Ordinary Differential Equations
- MA3252 Linear and Network Optimization
- MA3259 Mathematical Methods in Genomics
- MA3236 Nonlinear Programming
- MA3264 Mathematical Modeling
- MA3265 Introduction to Number Theory
- etc

Math with Specialization

5 additional modules at 4000 level

Pure Mathematics

MA4203 Galois Theory
MA4207 Mathematical Logic
MA4221 Partial Differential Equations
MA4229 Fourier Analysis and Approximation
MA4262 Measure and Integration
MA4271 Differential Geometry in Curves and Surfaces
MA4273 Algebraic Geometry in Curves and Surfaces

Operations Research and Analytics

MA4235 Graph Theory
MA4251/ST4238 Stochastic Processes II
MA4254 Discrete Optimisation
MA4260 Stochastic Operations Research
MA4264 Game Theory
MA4270 Data Modelling & Computation

Data Modelling and Analytics

MA4229 Fourier Analysis & Approximation
MA4230 Matrix Computation
MA4255 Numerical Methods in Differential Equations
MA4261 Coding and Cryptography
MA4268 Mathematics for Visual Data Processing
MA4270 Data Modelling & Computation

Combined Programmes

DDP and DMP

- Double Degree Programme:

Computer Science and MA

- Double Major Programmes:

MA + Second Major

Management (Biz)

Business Analytic (SoC)

Computer Science (SoC)

Information Security (SoC)

Many other DDP/DMP
combinations

Special Programme in Mathematics



- For selected students with strong passion and aptitude in mathematics.
- Provides in-depth study and rigorous training through specially designed modules and close mentorship.
- Students will build a firm foundation to pursue graduate programmes and future careers in the mathematical sciences
- Apply from Semester 2 of the first academic year

Blended Learning at NUS

Pedagogical advantage

*“evidence shows that it is not the technology per se that changes learning and teaching but the **pedagogical advantage** we make of its use”*

Price and Kirkwood (2008)

Price, L., and A. Kirkwood (2008) *Technology in the United Kingdom's higher education context*. In S. Scott & K. Dixon (Eds.), *The 21st Century, Globalised University: Trends and Development in Teaching and Learning* (pp. 83–113). Perth: Black Swan.

Key Elements of Blended Learning

1. Provide an opportunity for students to gain **first exposure** prior to class;
2. Provide an **incentive** for students **to prepare** for class;
3. Provide a **mechanism** to **assess student understanding**; and,
4. Provide **in-class activities** that focus on **higher level cognitive activities**.

Brame, C., (2013) *Flipping the classroom*, Vanderbilt University Center for Teaching [Retrieved 12 February 2015 from <http://cft.vanderbilt.edu/guides-sub-pages/flipping-the-classroom/>].

Using Multimedia for E-learning

Reducing extraneous processing

Coherence

Exclude interesting but irrelevant material as this reduces cognitive capacity to process essential material in a lesson.

Signaling

Include vocal cues and/or visual highlights to aid the selection & organization of important info, especially for learners with low prior knowledge.

Redundancy

Graphics with narration alone is more effective than also including on-screen text. Adding one or two keywords as on-screen text has benefit.

Contiguity

Place printed words near any corresponding graphics and coincide narration with related display.

Managing essential processing

Segmenting

Add self-pacing options to enable learners to process information before continuing.

Pre-training

Provide option to view information on key terms to allow learners to familiarize before having to work with them.

Modality

Present information about a graphic verbally rather than as text so that learners can listen and refer to graphic, especially for system paced dynamic graphics (e.g. videos).

Fostering generative processing

Personalisation

Present words in conversational style rather than formal style, including the use of personal pronouns (I & you) in script, especially in early stages

Voice

Narration should use a human voice rather than a computer voice, and this should match any on-screen character.

Embodiment

Drawing graphics as you explain is more beneficial than explaining a presented drawing as it reflects a real-life social interaction

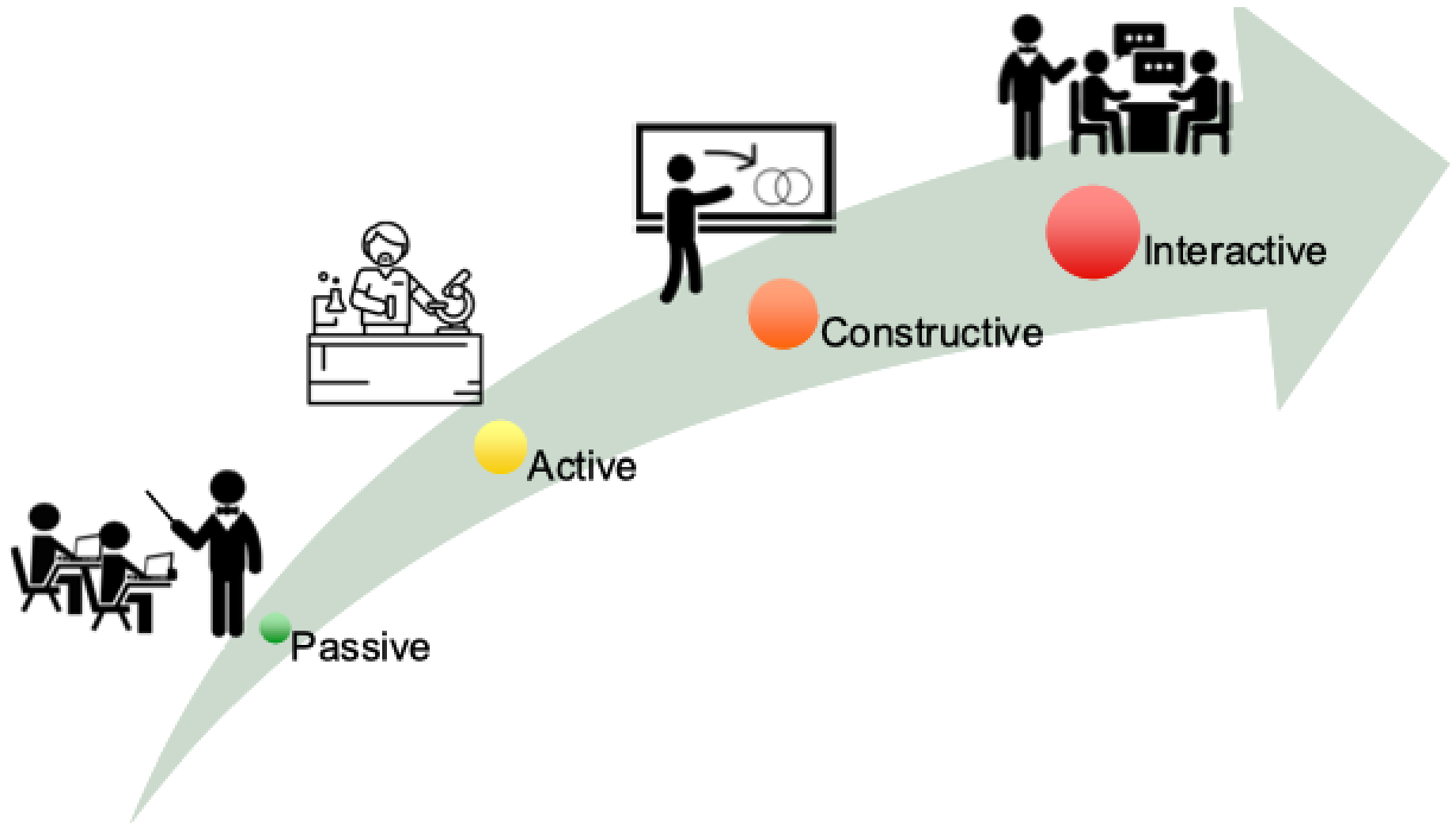
Just-in-Time Teaching

“The most important single factor influencing learning is what the learner already knows. Ascertain this and teach [them] accordingly.”

Ausubel (1968)

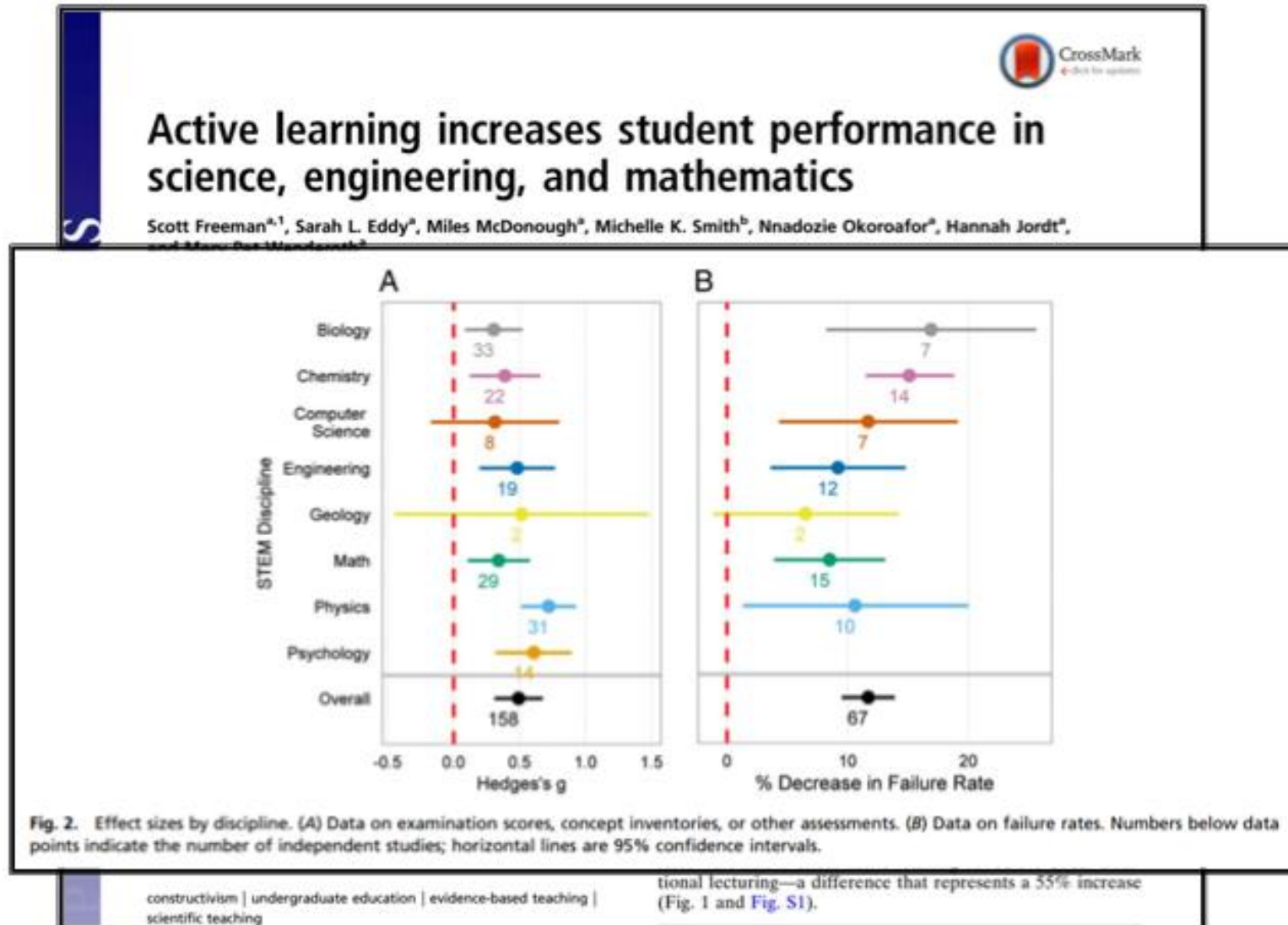
Novak, G. N., E. T. Patterson, A. Gavrin, and W. Christian (1999) *Just-in-Time Teaching: Blending Active Learning and Web Technology*, Saddle River: Prentice Hall.

Designing Learning Activities



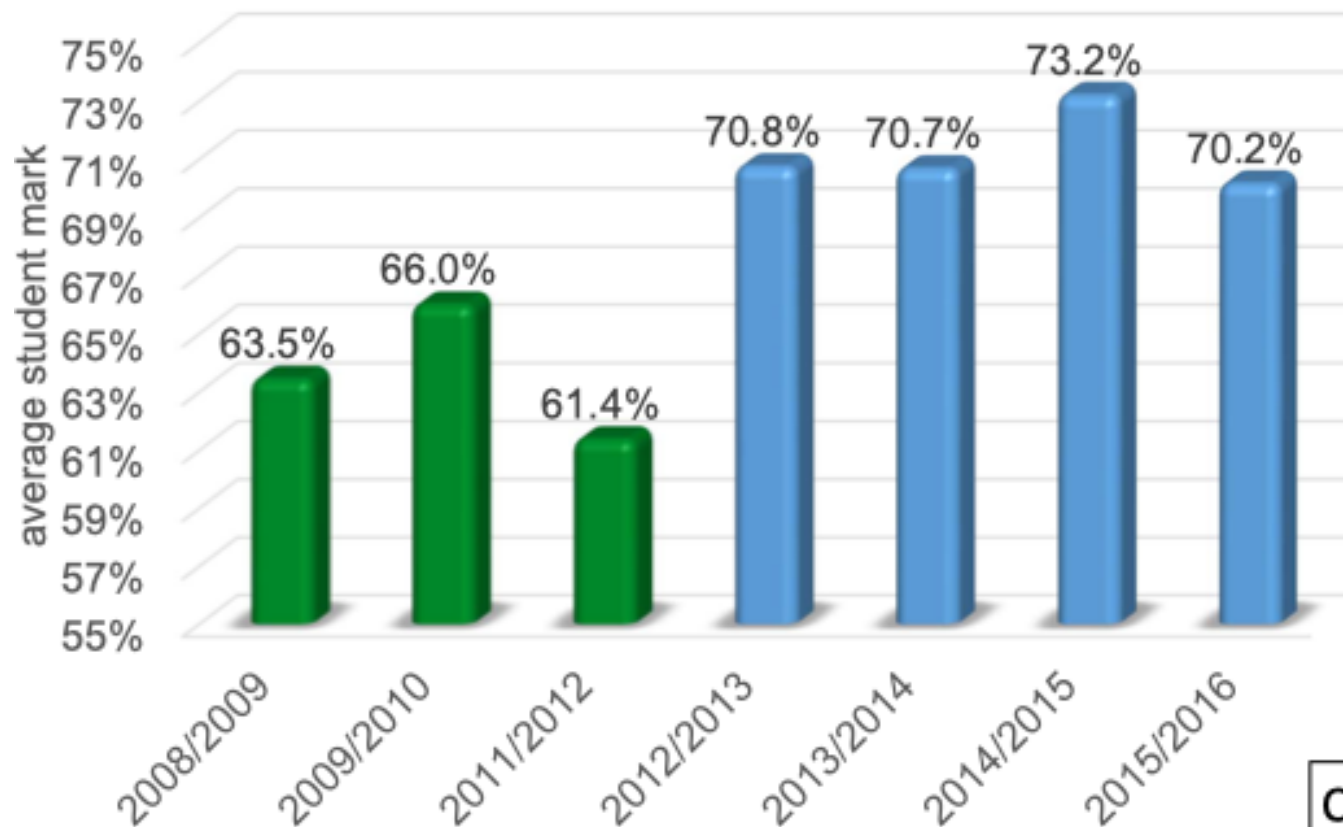
Mirriahi, N., D. Alonzo and B. Fox. (2015) A blended learning framework for curriculum design and professional development, *Research In Learning Technology*, 23: doi:<http://dx.doi.org/10.3402/rlt.v23.28451>.

Outcomes



Freeman, S., S. L. Eddy, M. McDonough, M. K. Smith, N. Okoroafor, H. Jordt and M. P. Wenderoth. (2014) Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*. 111, 8410–8415.

Student learning gain



Cohen's $d = 0.59$
 $p \text{ value} < 10^{-16}$

Reflecting on the learning process

“A ‘metacognitive’ approach to instruction can help students learn to take control of their own learning by defining learning goals and monitoring their progress in achieving them”

Bransford et al. (2000, p. 18)

Bransford, J. D., A. L. Brown and R. R. Cocking (2000) *How people learn: Brain, mind, experience, and school*. Washington, D.C.: National Academy Press.

Faculty and student
resistance to change

Rethinking the practice of teaching

“Without sufficient access to sustained support and the tools and resources essential in the design of a student-centered environment, instructors are challenged to create these experiences on their own. Managing the changing practice of teaching requires that institutions intentionally design faculty support that is not bound by location or time.”

Alexander et al. (2019, p. 19)

Alexander, B., K. Ashford-Rowe, N. Barajas-Murphy, G. Dobbin, J. Knott, M. McCormack, J. Pomerantz, R. Seilhamer and N. Weber (2019) *EDUCAUSE Horizon Report: 2019 Higher Education Edition*. Louisville, Co: EDUCAUSE.

Conclusions

- Mathematics is a powerful discipline that opens doors to science, technology, finance, education, research, and leadership.
- The mathematics programme at NUS is designed to provide both rigor and flexibility, helping students build strong foundations while exploring different pathways.
- In teaching mathematics, our goal is not only to transmit knowledge, but also to develop problem-solving ability, quantitative reasoning, and lifelong learning skills.

Conclusions

- Blended learning and thoughtful activity design create more space for active learning, reflection, and deeper understanding.
- Ultimately, teaching and learning mathematics in the technological era should empower students to use mathematics confidently, creatively, and responsibly in the world around them.